

INNOVATIVE MINING SYSTEMS

Dozer push mining

By

**Prof. Santosh Kumar Behera, Assistant Professor,
Mining Engineering**



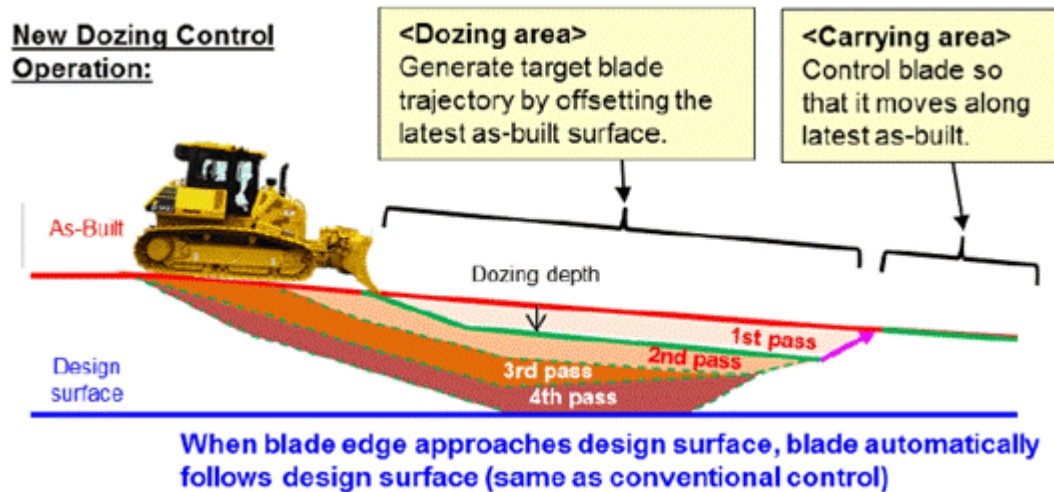
**Indian Institute of Technology (Indian
School of Mines), Dhanbad**

INTRODUCTION TO DOZER PUSH MINING

Dozer Push Mining is a surface mining method where high-horsepower crawler dozers push blasted overburden from the highwall towards a spoil dump instead of using conventional truck–shovel systems.

It is particularly suitable for:

- Shallow coal seams
- Flat to gently dipping deposits
- Soft to moderately hard overburden
- Short push distances (50–100 m)



Idealized Cast Blast / Dozer / Excavator PreStripping Range Diagram

Not To Scale

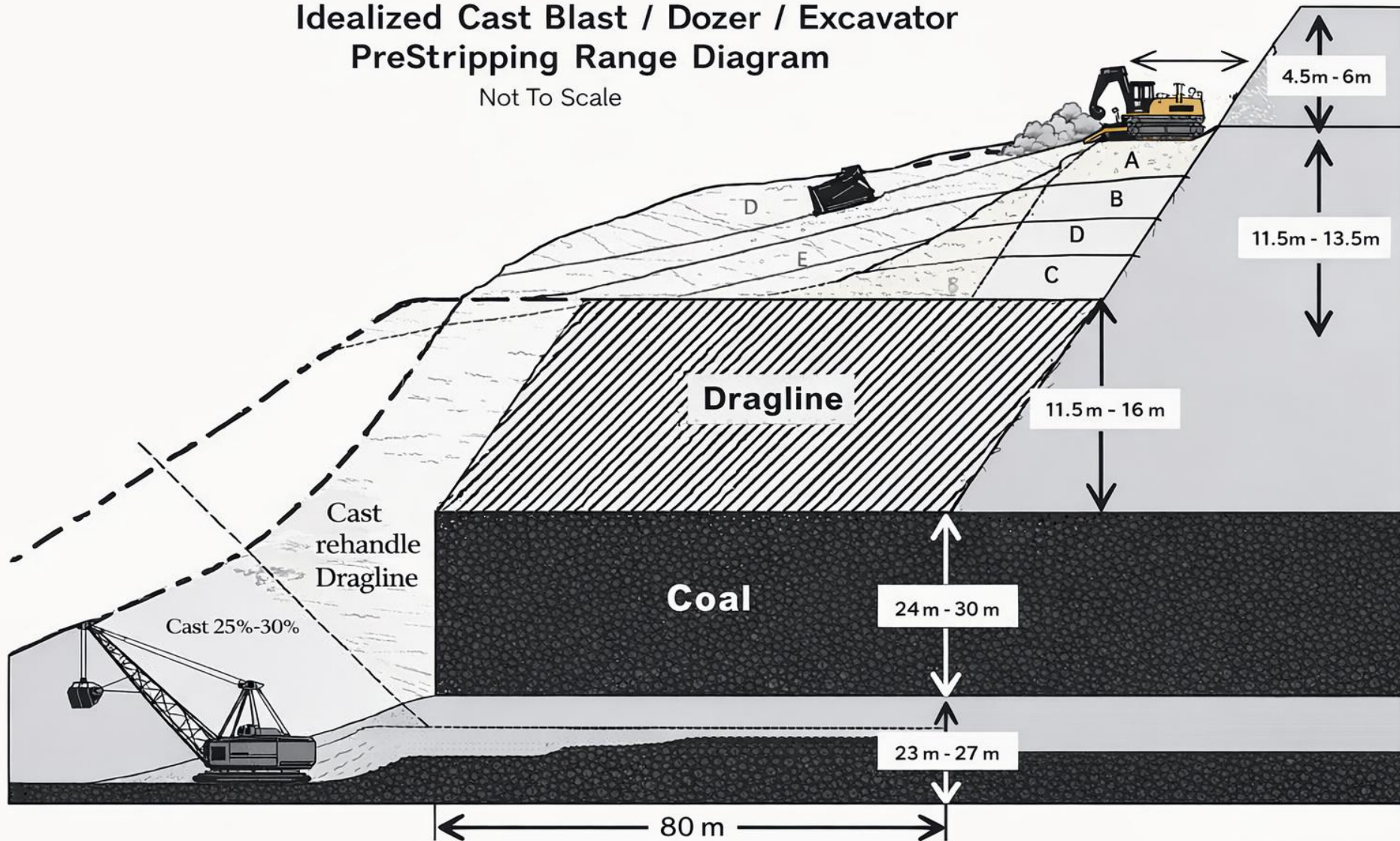
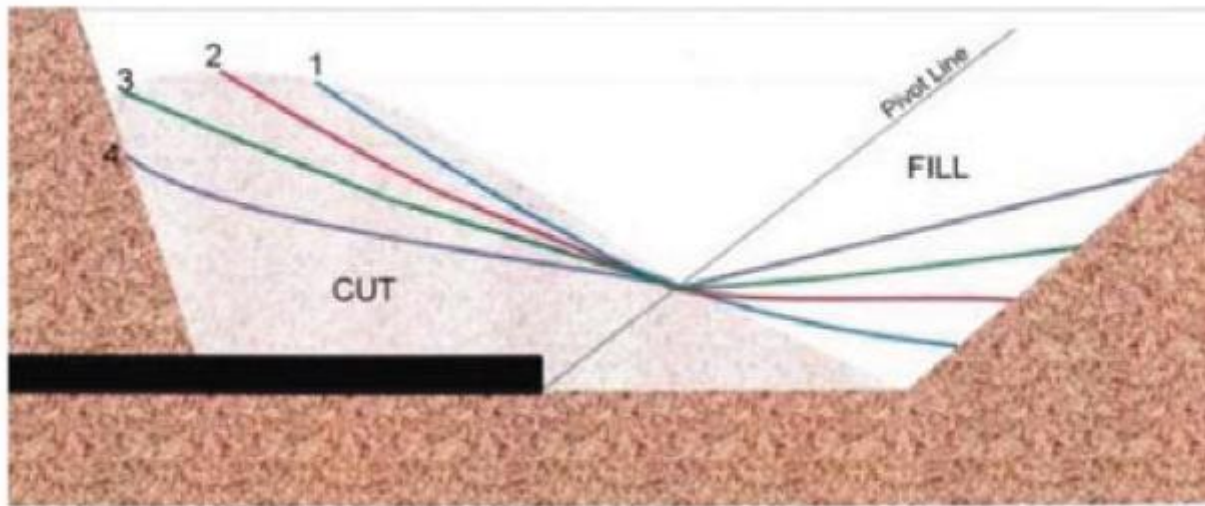


Figure 1—A typical dozer bulk push bench profile (Ingle et al., 2005)

DOZER PUSH BENCH GEOMETRY

Parameter	Typical Range
Bench width	45–60 m
Bench height	30–40 m
Highwall angle	~70°
Push distance	50–100 m



Pivot Point

The pivot point:

- Is drawn from toe of coal seam at $\sim 53^\circ$
- Separates CUT and FILL regions
- Material before pivot = Cut
- Material beyond pivot = Fill

Purpose:

- Maintain clean coal edge
- Optimize dragline performance
- Minimize rehandling

After blasting:

Material remains resting against highwall.

Dozers cannot:

- Get behind this material
- Start pushing effectively

Two solutions exist (from):

- Side-casts 12 m away
- Dozers reverse and push

- Cuts material
- Rotates blade
- Starts bulk push

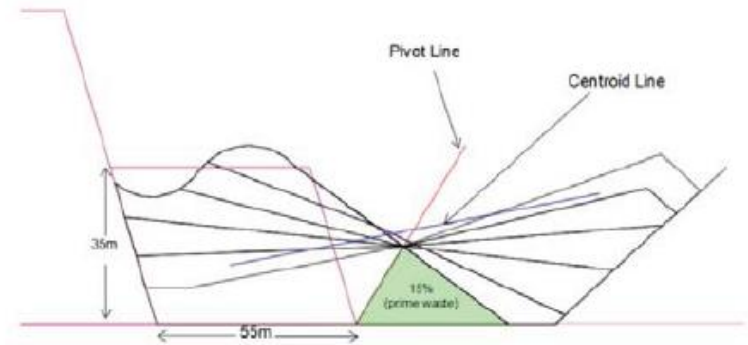


Figure 7—Small muckpile profile design without excavator assistance on 55 m bench

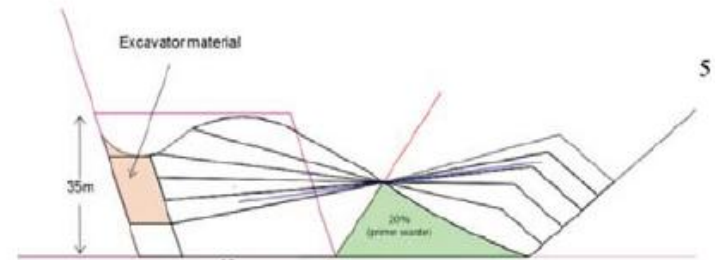


Figure 8—Medium muckpile profile design without excavator assistance on 55 m bench

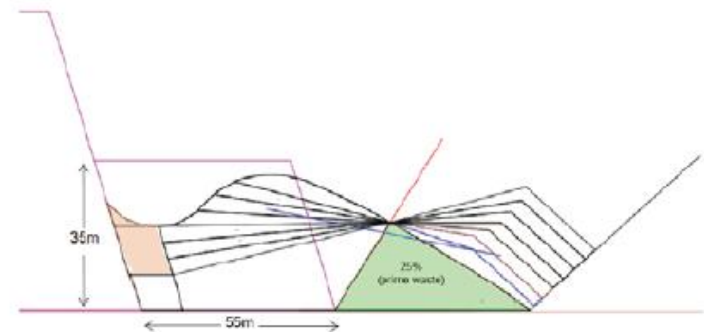


Figure 9—Large muckpile profile design without excavator assistance on 55 m bench

$$Productivity = Thp \times TF \times DoF \times OF \times DeF \times MF \times FF \times SF$$

$$Total\ Cost\ (A\$) = \sum_i^n \frac{Volume\ (LCM)}{Productivity\ (LCM/h)} \times Total\ Cost\ per\ Operating\ Hour$$

[2]

where:

n = number of slices per design scenario

Symbol	Meaning
Thp	Theoretical productivity
TF	Traction factor
DoF	Dozing factor
OF	Operator factor
FF	Fragmentation factor
SF	Side-cutting factor

Fragmentation	Fragmentation factor
Fine	1.0
Medium	0.85
Coarse	0.7

ADVANTAGES OF DOZER PUSH

- Lower capital cost
- Lower operating cost
- Reduced environmental disturbance
- Better monsoon operation
- Improved dragline efficiency
- Lower economic risk

DISADVANTAGES

- Cannot selectively mine
- High operator fatigue
- Highwall cutting difficulty
- Geological risk
- Limited to certain geology



PEKB(Parsa East & Kanta Basan) Opencast Coal Mine, Chattisgarh, ADANI.

- Designing Deep Hole Cast Blasting the project has successfully tailored the Dozer Push Mining Method for Indian coal mines.
 - The developed Dozer Push Mining introduces a paradigm shift by utilizing
 - unmanned, automated machinery for mining operations.
 - The drilling of 108 holes using automated drill machine (man-less), followed by cast/throw blasting using 60 tons of bulk emulsion explosives.
 - Further, the blasted material will be pushed in decoaled area using specially designed, large-sized automated dozer machine.
 - Improved dragline utilisation
 - Reduced monsoon delays
 - Enhanced safety due to unmanned operation
 - This marks a paradigm shift in Indian coal mining.
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- The dozer push mining method offers a viable alternative to the conventional truckshovel mining technique or shovel-dumper and dragline methods, with its distinct advantages such as improved efficiency by enabling faster coal recovery, reducing delays caused by adverse weather conditions such as the monsoon season and highly cost-effective, with an estimated 7-10% reduction in operational costs compared to conventional methods.
 - Additionally, it enhances productivity by improving the utilization of dragline machines and reducing unit costs. Moreover, the unmanned operation of the Dozer Push Mining method greatly enhances worker safety, minimizing the risks associated with manual labor and traditional mining techniques.

18. A dozer is working with the following parameters:

- $Thp = 1100 \text{ m}^3/\text{hr}$
- $TF = 0.9$
- $DoF = 0.95$
- $OF = 0.92$
- $DeF = 0.98$
- $MF = 0.9$
- $FF = 0.85$
- $SF = 0.75$

(a) Calculate the actual productivity.

(b) If fragmentation changes to coarse ($FF = 0.7$), calculate the new productivity.

(c) comment on the percentage of variation for productivity

Answer: 548, 452, Percentage Reduction=17.65%

INPUT DATA		
Theoretical Productivity (Thp)		1400
Traction Factor (TF)		0.88
Dozing Factor (DoF)		0.94
Operator Factor (OF)		0.9
Density Factor (DeF)		0.97
Material Factor (MF)		0.92
Medium Fragmentation (60%)		0.85
Coarse Fragmentation (40%)		0.7
Side Cutting Time (%)		0.3
Working Hours/Shift		8
Shifts/Day		2
Working Days/Year		300
Availability		0.85
Operating Cost per Hour (₹)		65000
Selling Price per m ³ (₹)		4000
Effective Side Cutting Factor		0.7
Weighted Fragmentation Factor		0.79
Actual Productivity per Dozer (m ³ /hr)		514
Effective Working Hours per Year		4080
Annual Production per Dozer (m ³)		2098582
Annual Production (3 Dozers) (m ³)		6295746
Total Annual Operating Cost (₹)		795600000
Gross Revenue (₹)		25182983036
Net Profit (₹)	₹	24,38,73,83,036
CASE: Push Distance = 100 m (12% Productivity Reduction)		
New Productivity per Dozer (m ³ /hr)		453
New Annual Production (3 Dozers) (m ³)		5540256
New Revenue (₹)		22161025071
New Profit (₹)	₹	21,36,54,25,071